Housing Price Prediction from Trulia.com Data

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12/06/21

Motivation

Suppose you want to buy a house in Denver. You find a property that you like, but you want to know if it's a good deal.

Now suppose you have a statistical model that predicts house prices in Denver. You plug in the features of your candidate house and get a price prediction back. You use this prediction to help determine if the house is property priced, overvalued, or undervalued.

The goal of our work in this project is to provide several candidate models for housing price prediction.

About the Data

Our data comes from Trulia.com and can be found on Kaggle[1]. It's a house listing website like Zillow.com. In fact it was acquired by Zillow in 2014 for $3.5 billion[2].

The dataset contains houses listed on the site from Sep 1, 2019 to Oct 31, 2019. For each house there are several features listed describing location, bedrooms, bathrooms etc.

Initially there were 30006 observations. Many features were not usable for prediction. These columns contained information such as "Unique House ID." After picking out the usable predictors and removing observations with missing information, the number of observations shrunk to 9999, which is still okay.

[1] https://www.kaggle.com/promptcloud/real-estate-data-from-trulia [2] https://money.cnn.com/2014/07/28/real_estate/zillow-buys-trulia-for-3-5-billion/index.html

Subsets

I elected to subset the data at both City and State levels. This was based on the assumption that housing markets tend to be more local than national.

For example I'm from Albuquerque, New Mexico where houses are quite expensive despite the low income levels of the population. This is starkly different from a place like Waco, Texas where houses are pretty cheap despite sometimes being quite large and otherwise Texas-y.

By subsetting our data into observations only from certain states and cities, I was able to create models with data that are more similar to those that will be used with the model.

The subset I chose were California, Los Angeles, Colorado, and Denver.

Problems with the Data

An issue we ran into in analysis was a lack of similar observations. For example in the California subset there was only one house with 36 bathrooms, so when it came to leaving out a test-set of our data if that observation was in the test-set, we couldn't actually predict based on bathroom since our model didn't contain a house with 36 bathrooms. The same thing is true for some numeric variables. Shrinkage methods however seemed to be robust against this issue. I'm not totally sure why.

Another analytical issue we ran into was perfect collinearity between dummy variables. When you have multiple categorical predictors weird things can happen where one column of your model matrix $\mathbf{X}$ is a linear combination of other columns and so least squares isn't possible, since $\mathbf{X}$ doesn't have an inverse. The only solution I found was to remove conflicting categorical predictors, which tended to be things that described the same information e.g. city and zipcode were sometimes perfectly collinear. Lack of observations was also probably a contributing factor to this issue.

Criteria for Linear Model Fitting

For each subset we made a 70%-30% training-testing split. This was because I wanted to test the models on observations in which it had not yet seen. For finding the best model parameters e.g. λ for Ridge Regression and the LASSO, I used 10-fold cross validation. Predictive model performance was based on MSE from the test set.

The types of models I fit were:

- Multiple Linear Regression (with all real predictors)
- Ridge Regression
- The LASSO
- Principal Component Regression
- Partial Least Squares Regression

Due to previously described issues, I didn't fit MLR with categorical predictors. Also, as far as I'm away it doesn't really make sense to fit PCR or PLSR with categorical predictors either, since taking a linear combination with dummy variables involved would be weird.

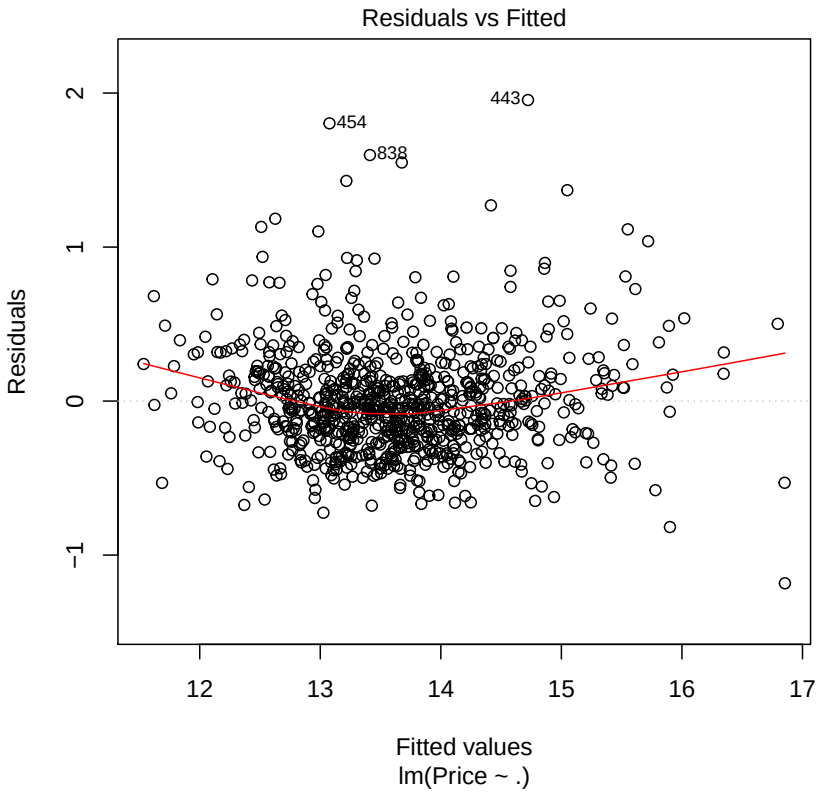
Data Transformations for Linear Models

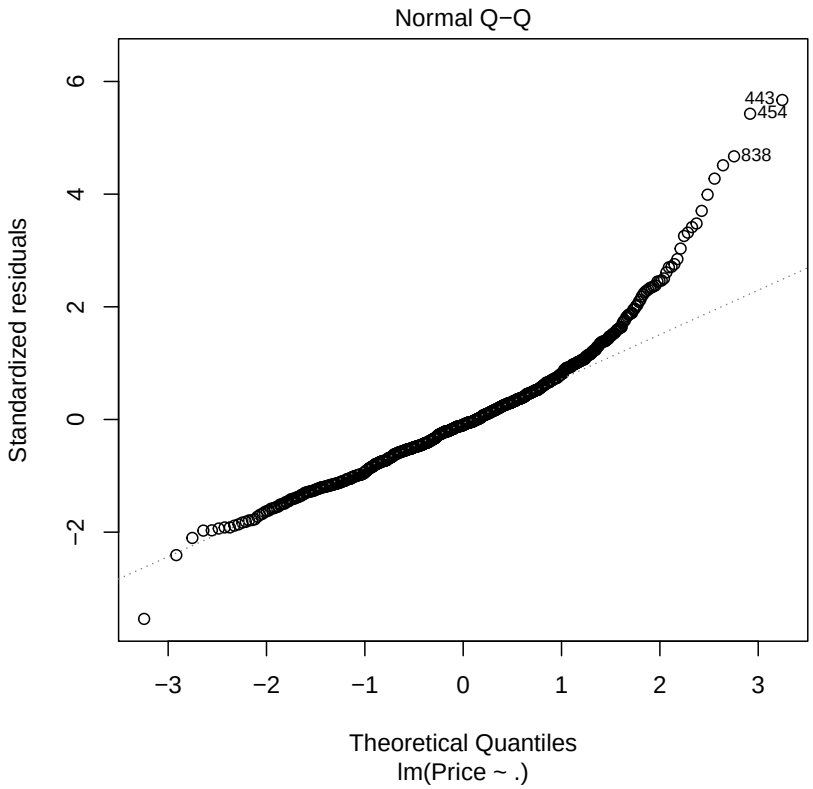
In order to fit linear regression models (and their generalizations) to the data, it was best to perform a few transformations first. This definitely has an impact on interpretability, but since our goal is prediction we chose not to care too much.

Most notably, for the linear models, we predicted the log(`Price`) instead of `Price`. Likewise I transformed some predictors by taking their log (mostly for those that represented prices e.g. last tax assessment, last sold for etc.), and others by taking their square-root.

This wasn't a perfect solution of course, but by fitting a linear regression model predicting log(`Price`) by all of the transformed predictors for the California subset we get the following diagnostic plots:

Data Transformations for Linear Models





Linear Models Results

California

The best performing model fit for the California subset was done with the LASSO:

```
## MSE (log(Price)):  0.08228367
```

```
## RMSE (after inverse transformation):  1359257
```

Colorado

The best performing model fit for the Colorado subset was done with Ridge Regression:

```
## MSE (log(Price):  0.03516985
```

```
## RMSE (after inverse transformation):  124241.8
```

Linear Models Results

Los Angeles

The best performing model fit for the Los Angeles subset was done with the LASSO:

```
## MSE (log(Price):  0.1532133
```

```
## RMSE (after inverse transformation):  1883764
```

Denver

The best performing model fit for the Denver subset was done with Ridge Regression:

```
## MSE (log(Price):  0.04000187
```

```
## RMSE (after inverse transformation):  151154
```

Linear Models Conclusions

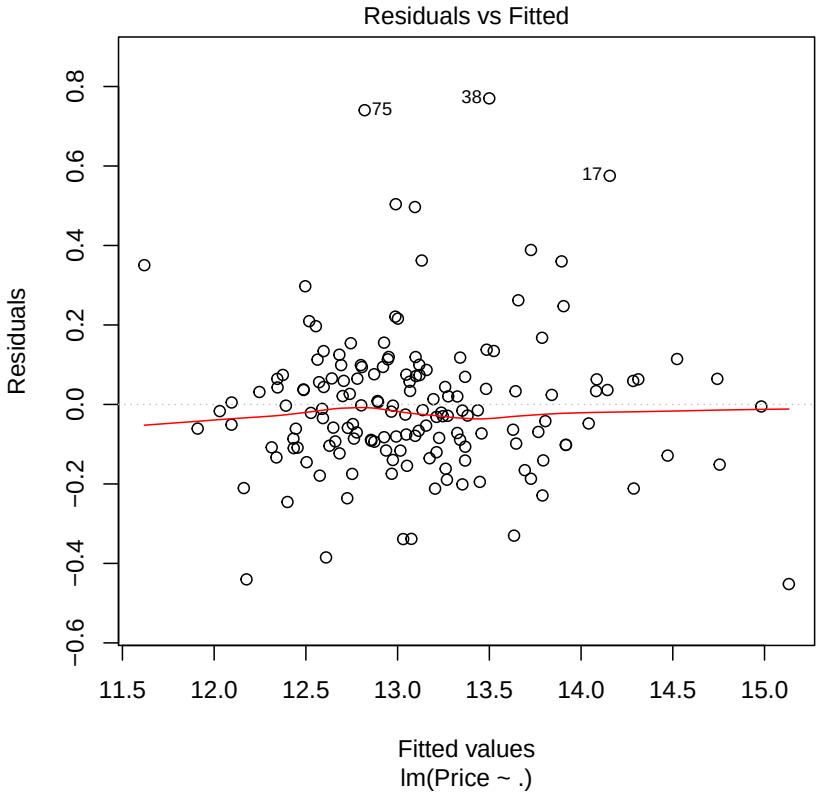
The main reason these test errors are so high is likely the lack of similar observations as previously described. One reason that the errors for Colorado and Denver are lower is probably due to the smaller range of prices listed for those two subsets:

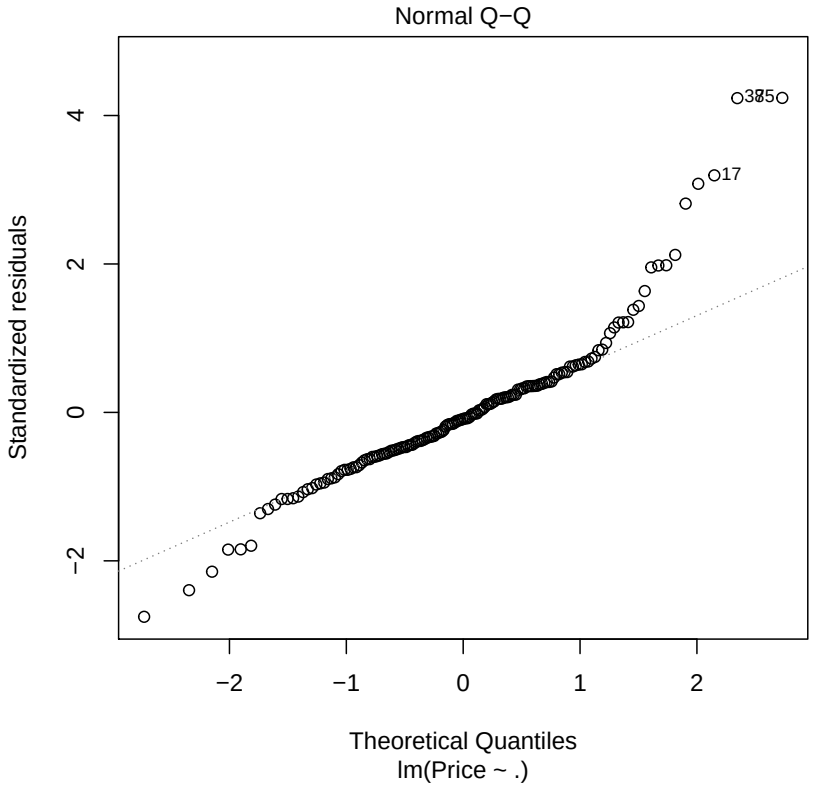
```
## Price Range CA:  70000 32500000

## Price Range CO:  108000 4e+06

## Price Range LA:  339000 18950000

## Price Range DEN:  108000 3195000
```

Not having enough observations is probably a difficult problem to overcome. I'm not sure if removing so much data at the start for having missing features was a smart move. Maybe I should have looked closer at what information was missing and decided to remove predictors rather than observations.

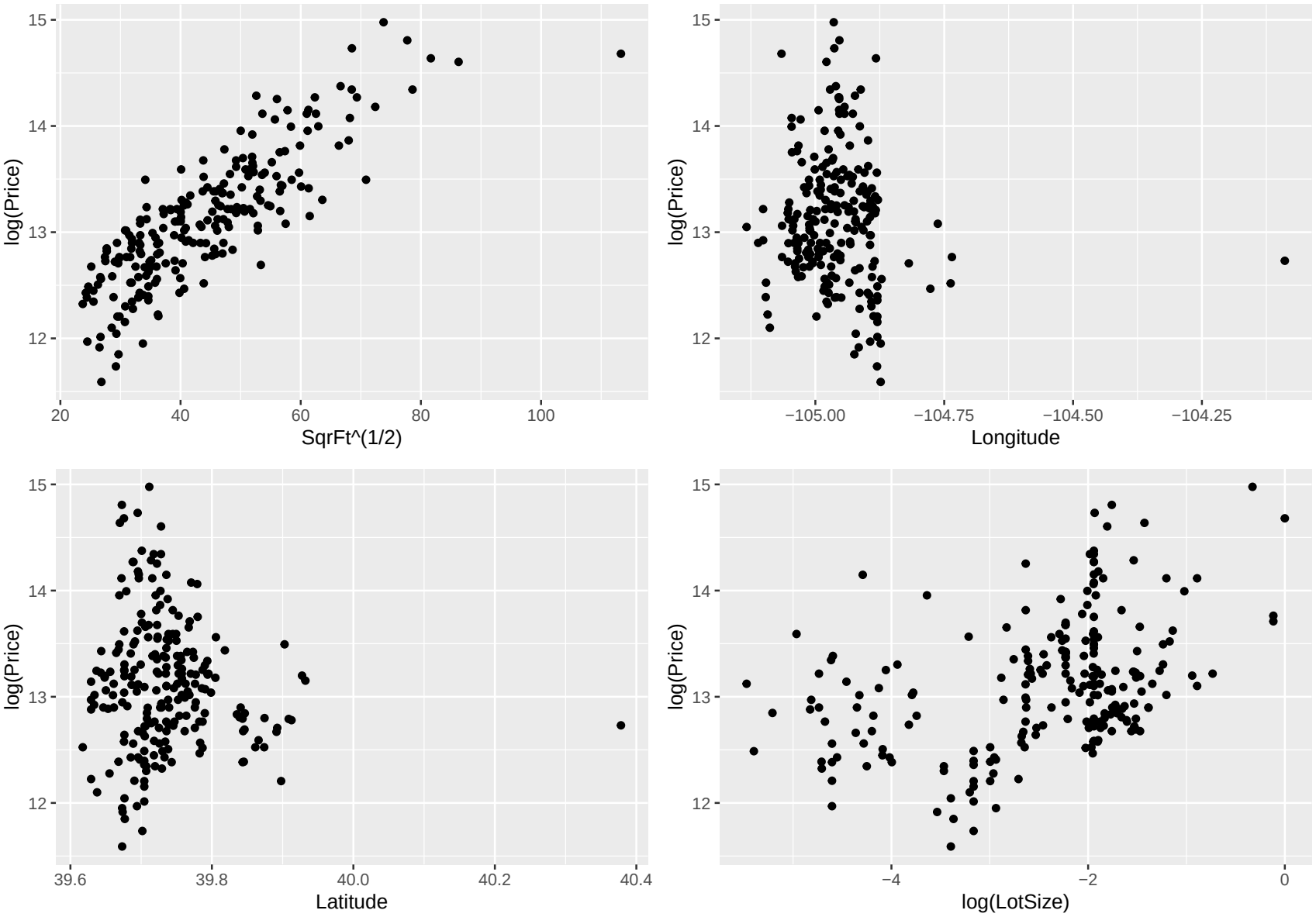
Linear Models Conclusions

Another reason could be that a linear fit worked better for that subset of data e.g. for Denver:

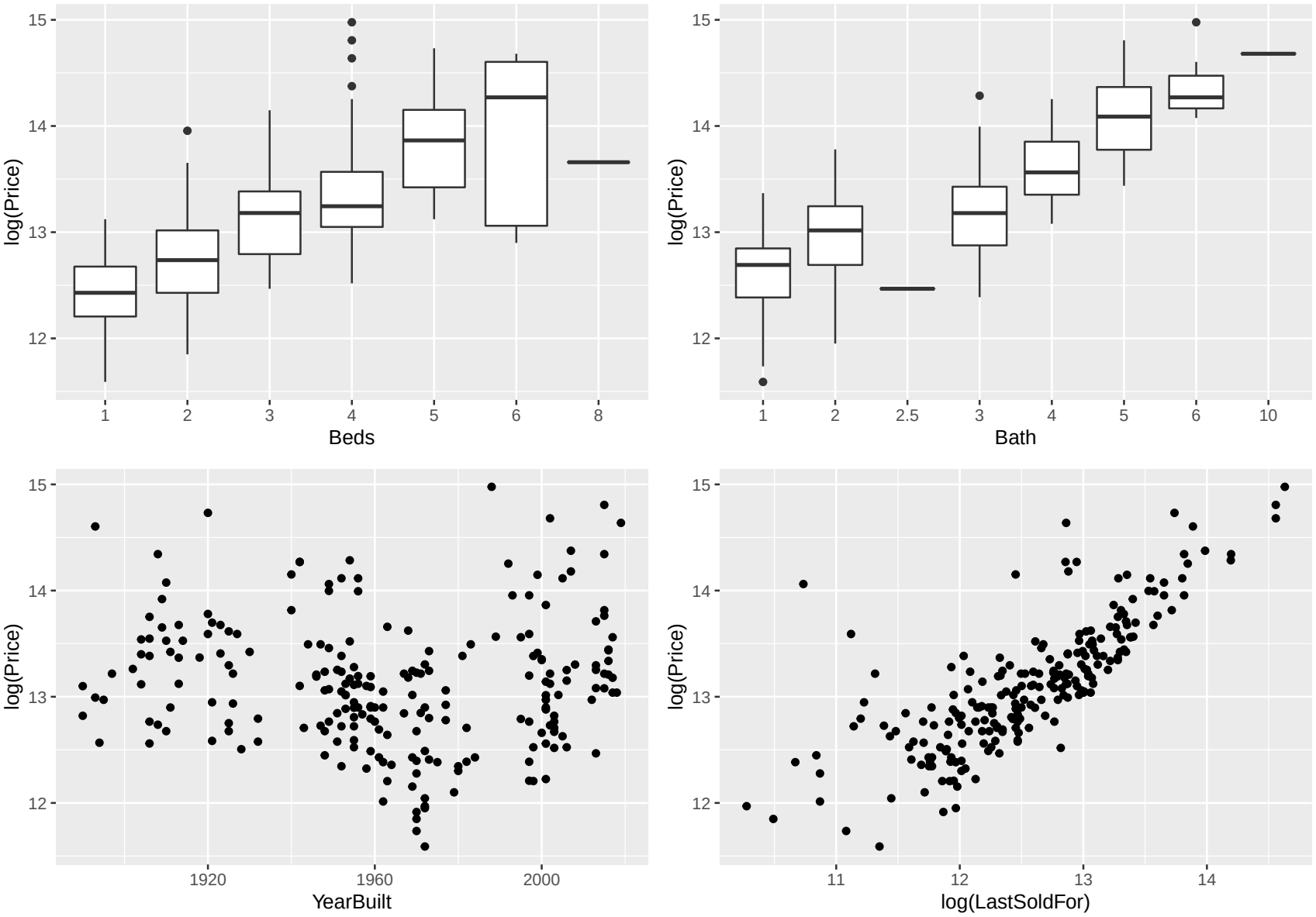




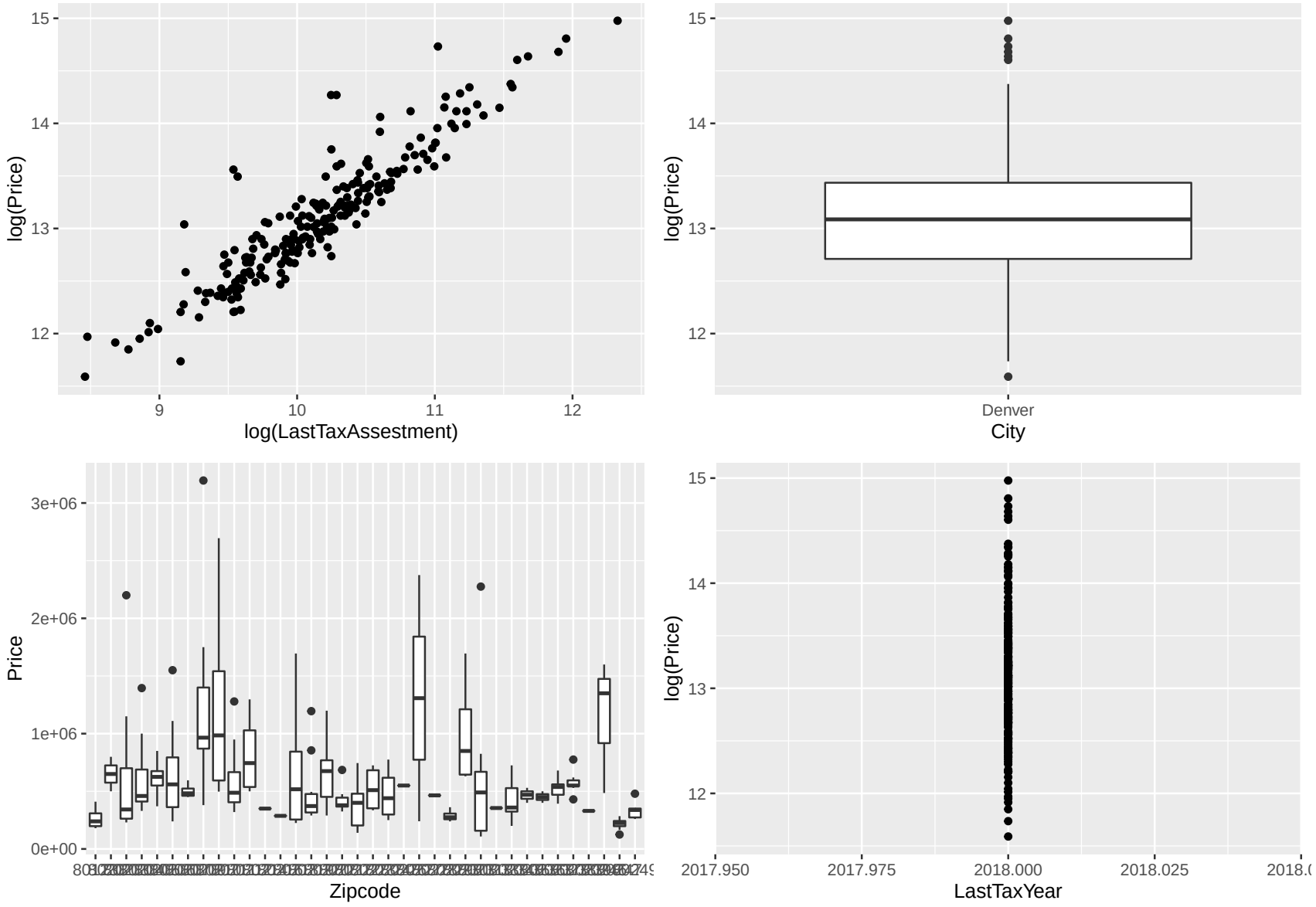
Denver subset data(1)



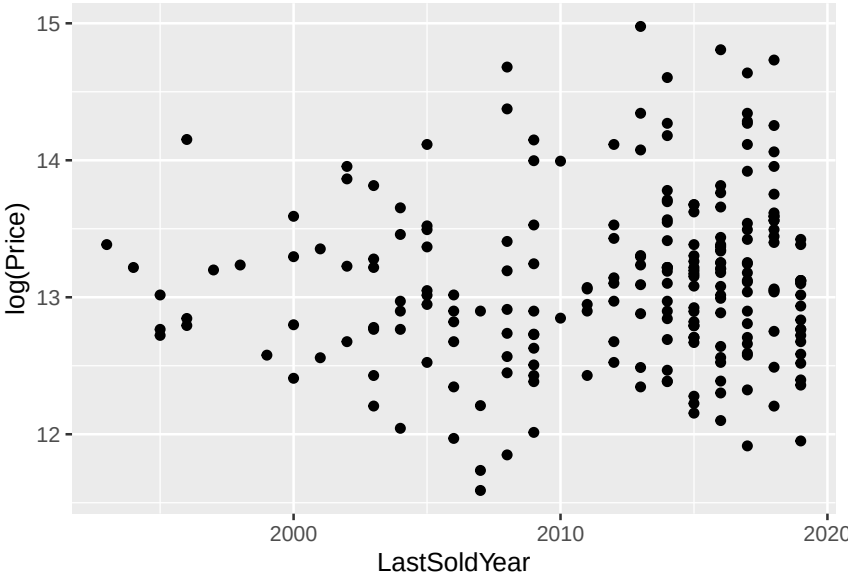
Denver subset data(2)

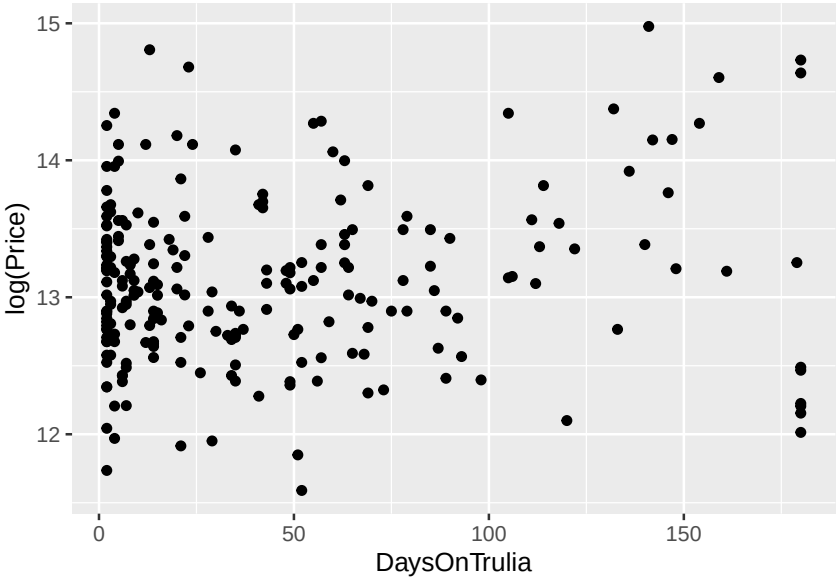


Denver subset data(3)

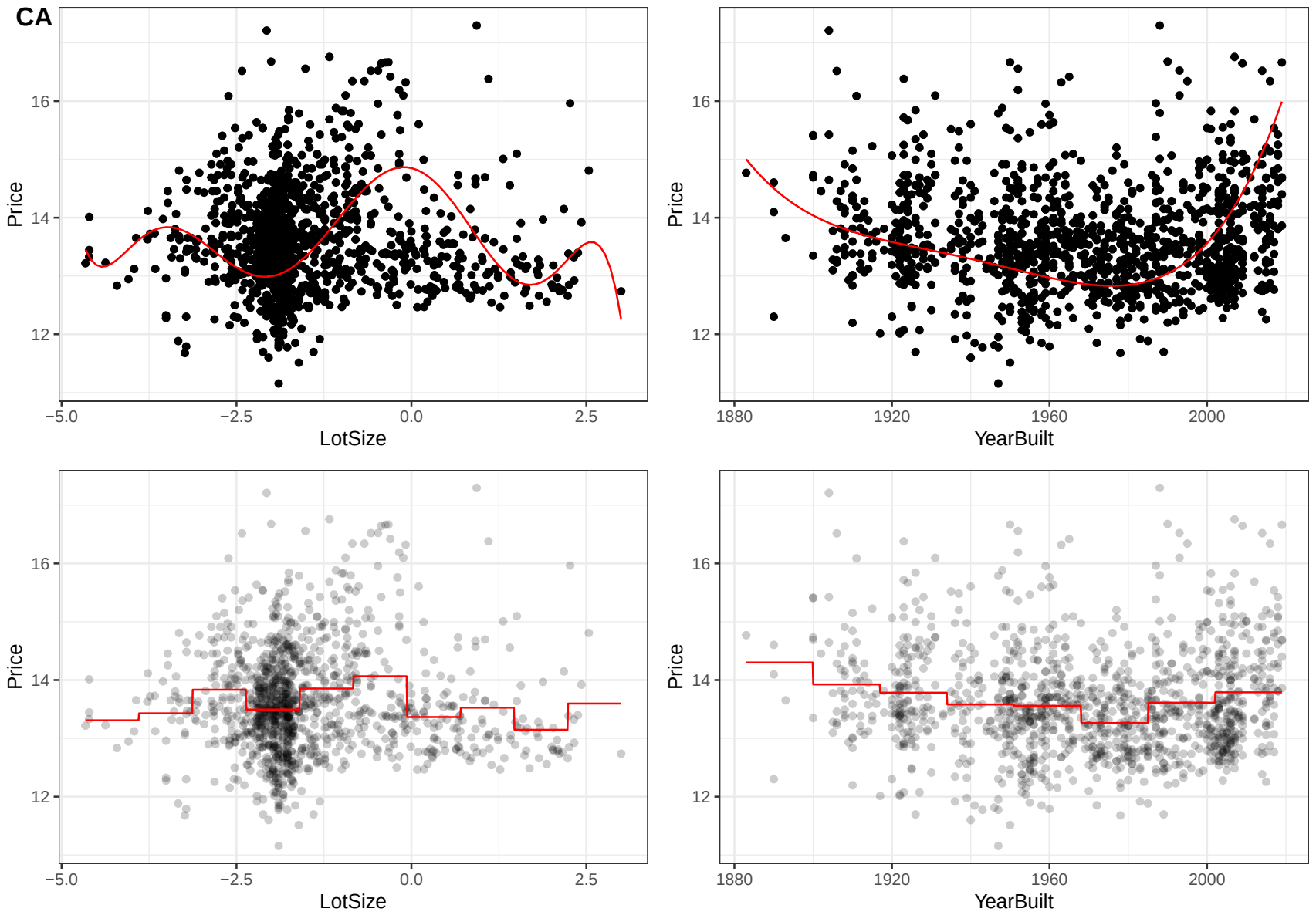


Denver subset data(4)

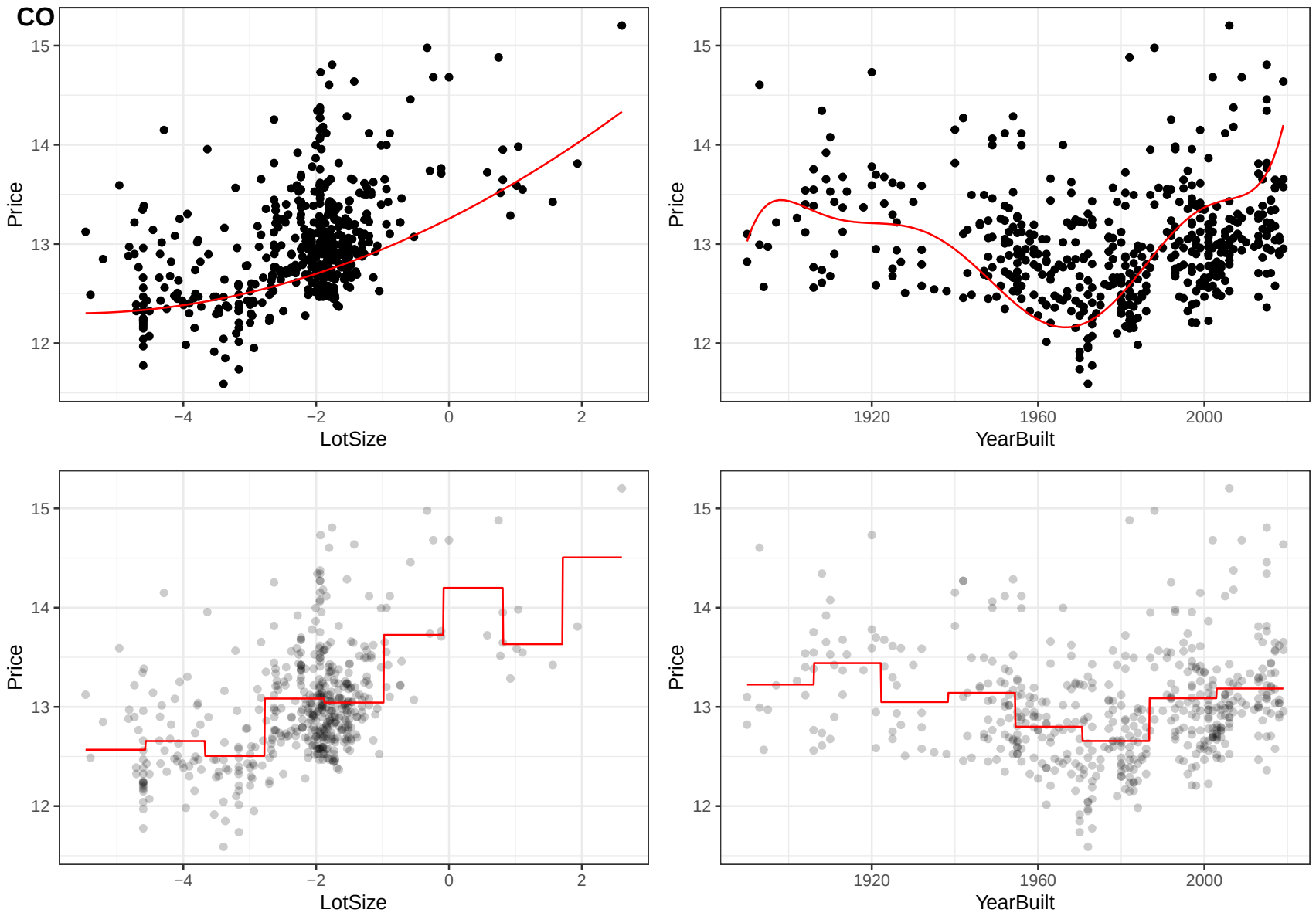




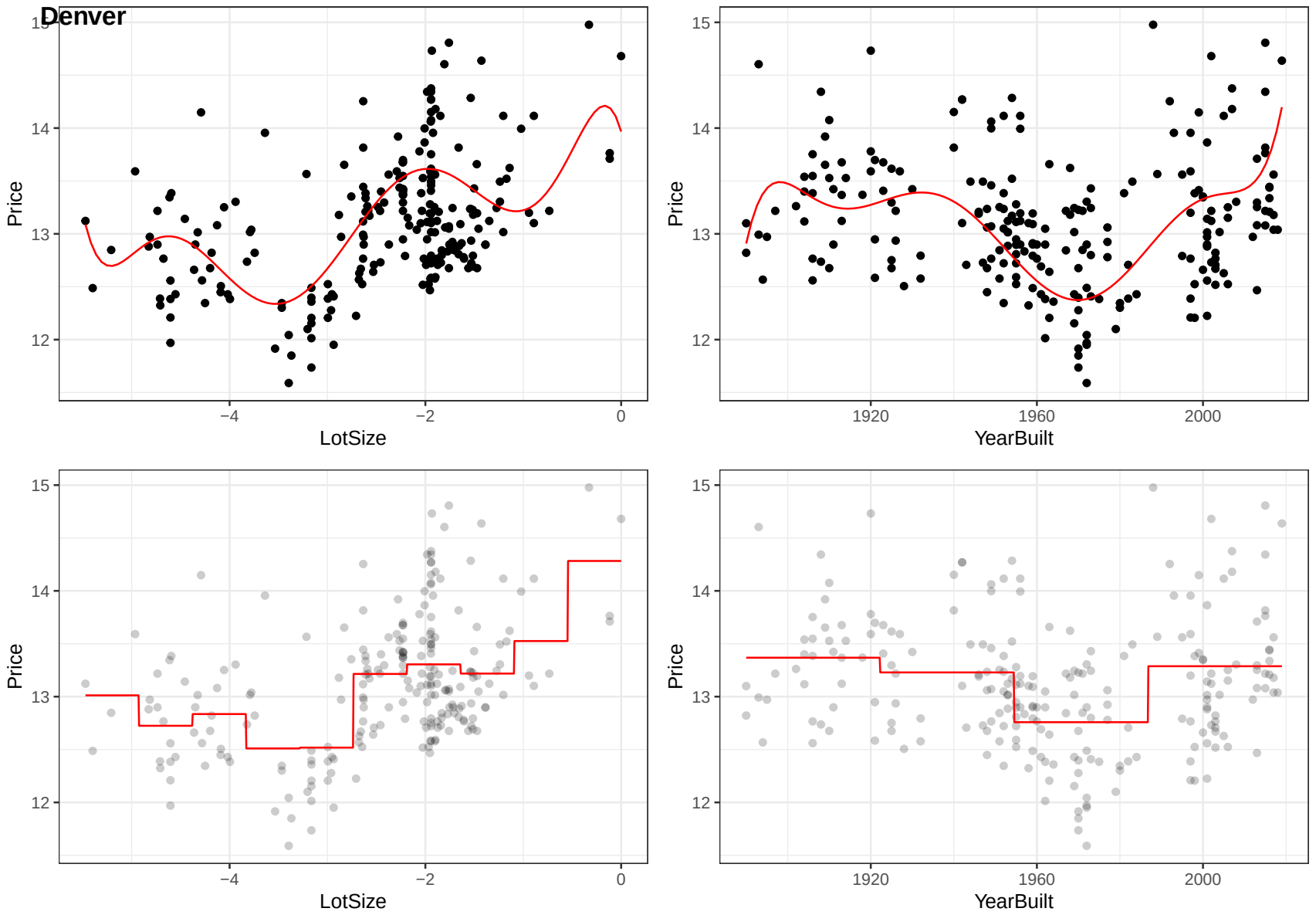
Polynomial regression



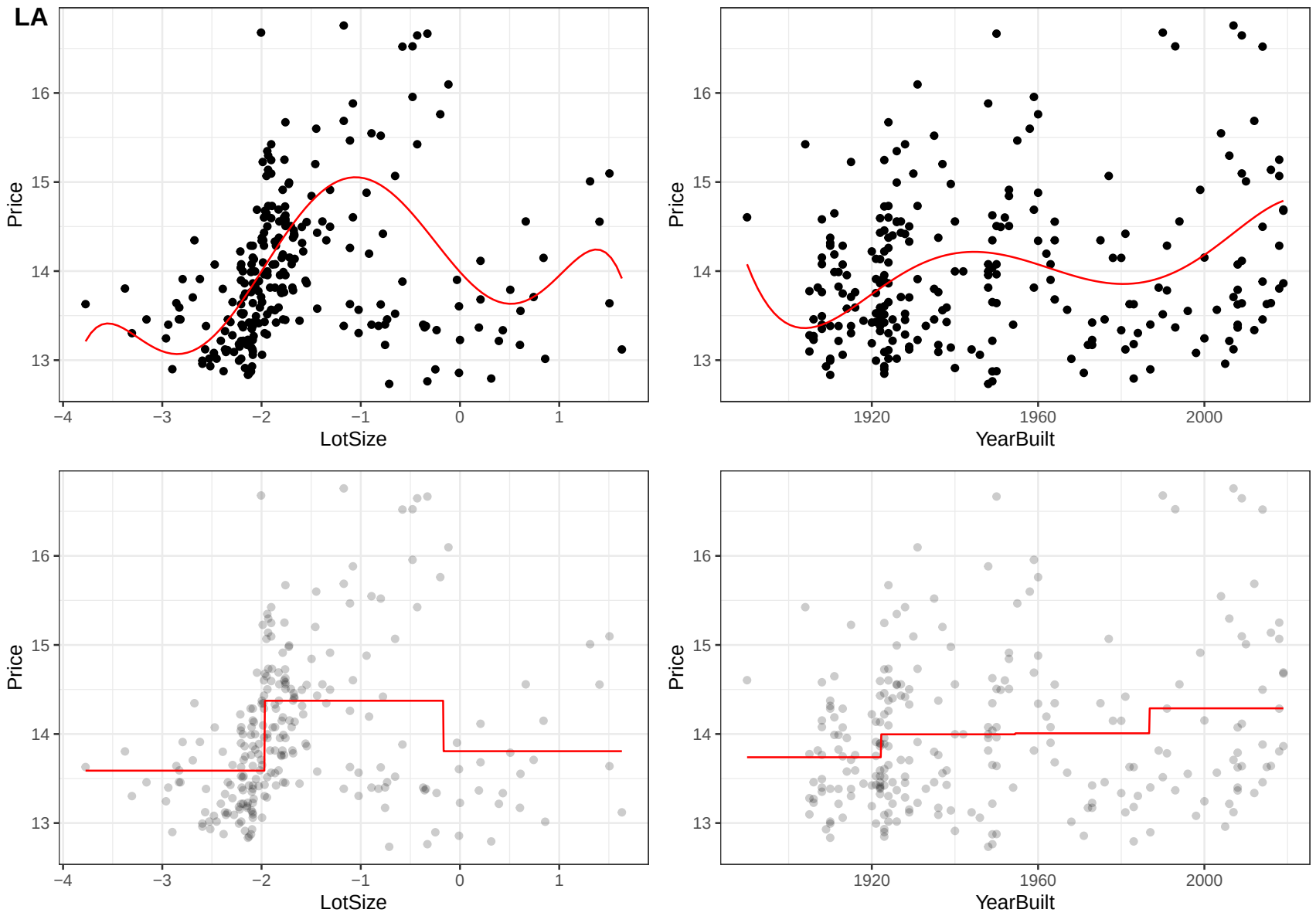
Polynomial regression



Polynomial regression



Polynomial regression



Polynomial result

```
## CA

## MSE (log(Price)) using Lot Size:  0.7692414

## MSE (log(Price)) using Year Built:  0.7737404

## CO

## MSE (log(Price)) using Lot Size:  0.2235432

## MSE (log(Price)) using Year Built:  0.2364161

## Denber

## MSE (log(Price)) using Lot Size:  0.2560896

## MSE (log(Price)) using Year Built:  0.3099499

## LA

## MSE (log(Price)) using Lot Size:  0.5387876
```

Slide 1

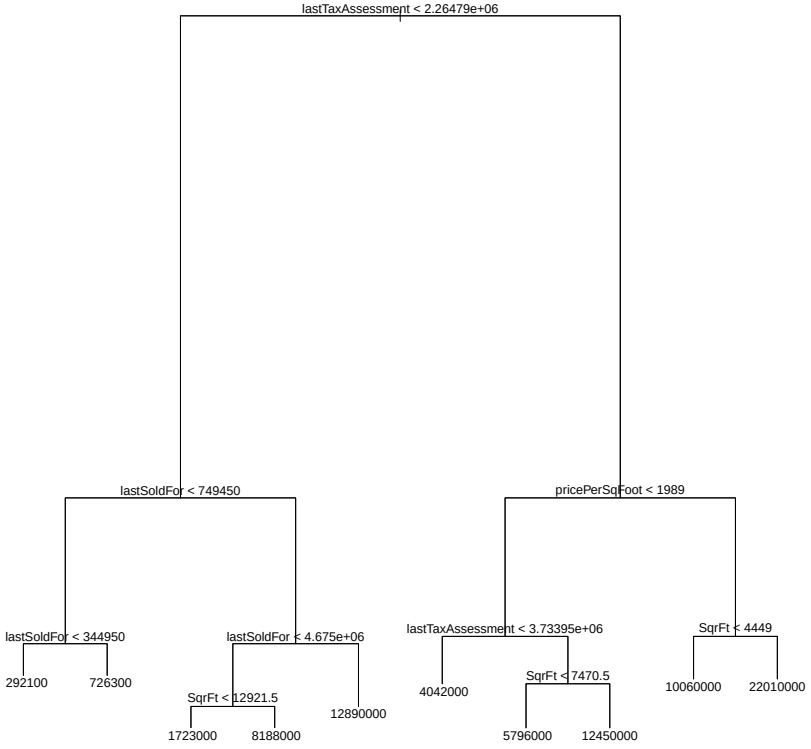
The focus of my slide will be trees, random forests, bagging, and boosting.

Decision tree finding ideal set of variables for regression model. Optimal model was lastTaxAssessment, lastSoldFor, SqrFt and pricePerSqrFt.

```
## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 9999 1.293e+16   575400
##    2) lastTaxAssessment < 2.26479e+06 9885
##      4) lastSoldFor < 749450 9150 1.146e+1
##        8) lastSoldFor < 344950 7004 4.006e
##        9) lastSoldFor > 344950 2146 4.358e
##      5) lastSoldFor > 749450 735 2.055e+15
##       10) lastSoldFor < 4.675e+06 730 9.62
##         20) SqrFt < 12921.5 725 6.091e+14
##         21) SqrFt > 12921.5 5 1.458e+14  8
```

Slide 2

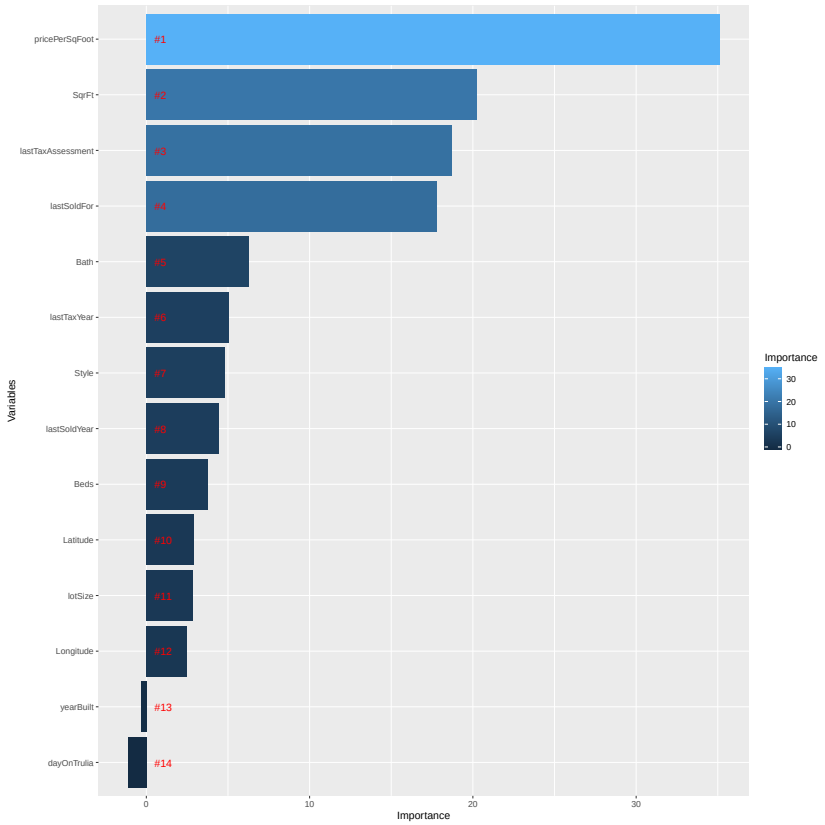
The following model as as follows: last tax assessment; last sold for; price per square foot; square footage.



Slide 3

First Random Forest model: No. of variables tried at each split: 4
Mean of squared residuals: 177222506973 % Var explained: 84.67

```
## [1] "Plot of variable importance"

## [1] "Variable importance of model"
```



Slide 4

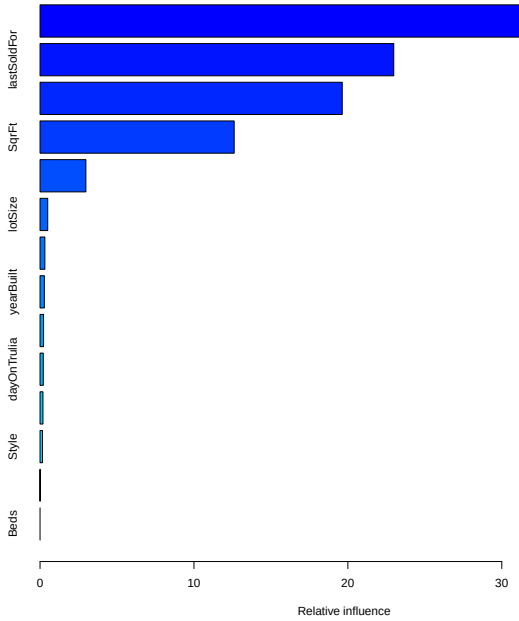
Bagging results: Number of trees: 500 No. of variables tried at each split: 14

```
Mean of squared residuals: 208185396457
          % Var explained: 81.99


##                 Length Class  Mode
## call               5   -none- call
## type               1   -none- character
## predicted       7999   -none- numeric
## mse              500   -none- numeric
## rsq              500   -none- numeric
## oob.times       7999   -none- numeric
## importance        28   -none- numeric
## importanceSD      14   -none- numeric
## localImportance    0   -none- NULL
## proximity          0   -none- NULL
## ntree              1   -none- numeric
```

Slide 5

Boosting model with 5000 trees and a depth of 4.
Variable:Relative Influence;

pricePerSqFoot : 33.414488750 lastTaxAssessment : 23.819157650
lastSoldFor : 20.898101638 SqrFt : 14.973841844




```
##                                   var      r
## pricePerSqFoot       pricePerSqFoot 39.913
## lastSoldFor             lastSoldFor 22.981
## lastTaxAssessment lastTaxAssessment 19.634
```

Slide 6

```
$Random Forest predictions:
```

278592.8 449497.0 339443.3 248671.0 217065.4 335604.2 1128607.5 285870.0

```
$Random Forest 2 predictions:
```

295631.6 459782.7 364393.7 252325.0 224685.8 326962.4 1122754.4 280512.3

```
$test_subset:
```

235288 230000 390000 109000 265000 239000 355000 259900

```
$Bag predictions:
```

265203.4 438926.6 301184.0 240096.8 197319.4 372527.2 1174993.5 284722.8

```
$Boost predictions:
```

273667.1 443304.5 298213.3 252108.5 189233.0 365708.5 1195343.4 285309.7